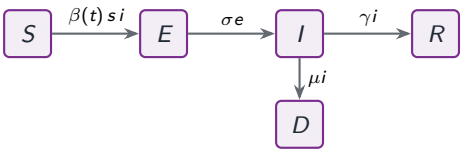
Why transmission must change over time

Module 1 ▪ Ebola in Sierra Leone, 2014–15 ▪ an SEIRD model with a time-varying transmission rate $\beta(t)$

The problem

- Ebola spreads by contact with body fluids, and with the dead at funerals
- isolation, safe burials and contact tracing cut transmission as the epidemic went on: β is **not** a constant
- surveillance reports only cases and deaths; S and E are never measured



$\dot{s} = -\beta s i, \quad \dot{e} = \beta s i - \sigma e, \quad \dot{i} = \sigma e - (\gamma + \mu) i, \quad \dot{r} = \gamma i, \quad \dot{d} = \mu i$

fractions of N, so $s + e + i + r + d = 1$ at all times

	Meaning	Value
N	population (2014 census)	7,092,113
σ	incubation rate (9.4 days)	0.1064 /day
γ	recovery rate (7 days)	0.1429 /day
μ	death rate	0.1000 /day
β	transmission rate (baseline)	0.4858 /day
CFR	$\mu/(\gamma + \mu)$	41.2%
R_0	$\beta/(\gamma + \mu)$	2.00

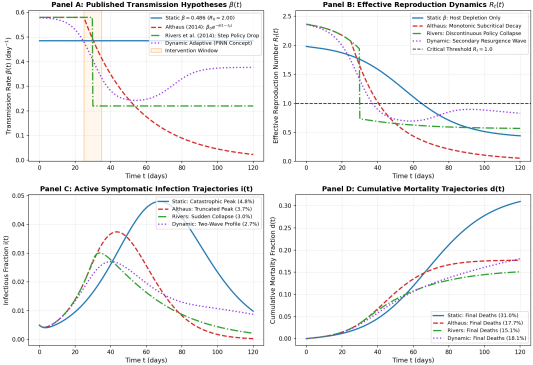
Assumptions

- closed population, homogeneous mixing
- recovered people stay immune
- σ, γ fixed from WHO clinical data, so the data only have to determine $\beta(t)$ and μ

Why $\beta(t)$: with a constant β ($R_0 = 2$) the model predicts about **80%** of Sierra Leone infected and **33%** dead. WHO confirmed 8,704 cases, about **0.12%**. The difference is that transmission fell.

Published models: how others handled $\beta(t)$

Module 1 ▪ each makes one assumption about how transmission changes ▪ figure: illustrative shapes over 120 days



Model	$\beta(t)$	Limitation
Chowell (2004)	drops after control starts	shape assumed
Legrand (2007)	3 routes: $\beta_I, \beta_H, \beta_F$	routes not separable
Rivers (2014)	step at a policy date	abrupt; date assumed
Althaus (2014)	$\beta_0 e^{-k(t-t_0)}$	can only fall
Our PINN	learned from data	smooth, any shape

Legrand: from one national case curve the three routes trade off against each other.

Althaus, in one line:

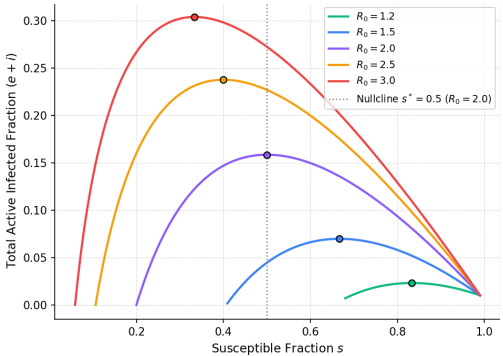
$$\frac{dR_t}{dt} = -\frac{\beta s}{\gamma + \mu}\left[k + \beta i\right] < 0$$

so once $R_t < 1$ it can never rise again.

The gap: every published model fixes the *shape* of $\beta(t)$ in advance (a date, a step, a decay). Our PINN learns $\beta(t)$ from cases and deaths, with no date or shape assumed.

Phase plane: the peak and the control threshold

Module 1 ▪ exact formulas, checked against an accurate ODE solver (DOP853)



$R_0 = 1.2$–3.0; time runs right to left; dots = predicted peaks

Divide the equations: i cancels exactly

$$\frac{d(e+i)}{ds} = -1 + \frac{1}{R_0\, s}$$

Peak where the slope is zero: $s^* = 1/R_0$

$$(e+i)_{\max} = (e_0 + i_0) + s_0 - \tfrac{1}{R_0}\left[1 + \ln(R_0 s_0)\right]$$

- $R_0 = 2$: peak **15.85%** at $s = 0.5$ (i alone about 4.8%)
- formula vs. ODE: relative error at most 1.3×10^{-5}

Control threshold

$$p_c = \delta_c = 1 - \frac{1}{R_0} = 50\% \quad (R_0 = 2)$$

What it means for control: without a vaccine, isolation, safe burials and tracing together must cut transmission by **more than 50%** to turn the epidemic. These exact formulas are also the checks used for the solvers (Module 3) and the PINN (Module 4).